Glossary

Aligned
A data item stored at an address that is divisible by the highest power of 2 that divides into its size in bytes. Aligned halfwords, words and doublewords therefore have addresses that are divisible by 2, 4 and 8 respectively.

An aligned access is one where the address of the access is aligned to the size of each element of the access.

At approximately the same time
Two events occur at approximately the same time if a remote observer might not be able to determine the order in which they occurred.

AXI beat
See Beat.

AXI burst
See Burst.

AXI transaction
See Transaction.

Barrier
An operation that forces a defined ordering of other actions.

Beat
An individual data transfer within an AXI burst.

See also Burst, Transaction.

Big-endian memory
Means that *the most significant byte* (MSB) of the data is stored in the memory location with the lowest address.

See also Endianness, Little-endian memory,

Blocking
Describes an operation that prevents following actions from continuing until the operation completes.

A non-blocking operation can permit following operations to continue before it completes.

Branch prediction
Is where a processor selects a future execution path to fetch along. For example, after a branch instruction, the processor can choose to speculatively fetch either the instruction following the branch or the instruction at the branch target.

See also Prefetching.

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Burst
In an AXI transaction, the payload data is transferred in a single burst, that can comprise multiple beats, or individual data transfers.

See also Beat, Transaction.

Byte
An 8-bit data item.

Cache
Any cache, buffer, or other storage structure in a caching master that can hold a copy of the data value for a particular address location.

Cache hit
A memory access that can be processed at high speed because the data it addresses is already in the cache.

Cache line
The basic unit of storage in a cache. Its size in words is always a power of two. A cache line must be aligned to the size of the cache line.

The size of the cache line is equivalent to the coherency granule.

See also Coherency granule.

Cache miss
A memory access that cannot be processed at high speed because the data it addresses is not in the cache.

Caching master
A master component that has a hardware-coherent cache. A caching master has a snoop address and snoop response channel, and optionally, a snoop data channel.

A master component might have only non-coherent caches. These caches can be for private data or they can be software-managed to ensure coherency. A master with a non-coherent cache is not a caching master. That is, the term *caching master* refers to a master with a cache that the ACE protocol must keep coherent.

See also *Initiating master*, *Master component*, *Snooped master*.

Coherent
Data accesses from a set of observers to a memory location are coherent accesses to that memory location by the members of the set of observers are consistent with there being a single total order of all writes to that memory location by all members of the set of observers.

Coherency granule
The minimum size of the block of memory affected by any coherency consideration. For example, an operation to make two copies of an address coherent makes the two copies of a block of memory coherent, where that block of memory is:

- at least the size of the coherency granule
- aligned to the size of the coherency granule.

In the ACE specification, the coherency granule is the cache line size.

See also Cache line.

Component
A distinct functional unit that has at least one AMBA interface. Component can be used as a general term for master, slave, peripheral, and interconnect components.

See also Interconnect component, Master component, Memory slave component, Peripheral slave component, Slave component.

Device
See Peripheral slave component.

Downstream
An AXI transaction operates between a master component and one or more slave components, and can pass through one or more intermediate components. At any intermediate component, for a given transaction, *downstream* means between that component and a destination slave component, and includes the destination slave component.

Downstream and upstream are defined relative to the transaction as a whole, not relative to individual data flows within the transaction.

See also Master component, Slave component, Upstream.

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Downstream cache

A downstream cache is defined from the perspective of an initiating master. A downstream cache for a master is one that it accesses using the fundamental AXI transaction channels. An initiating master can allocate cache lines into a downstream cache.

See also Downstream, Initiating master.

Deprecated

Something that is present in the specification for backwards compatibility. Whenever possible you must avoid using deprecated features. These features might not be present in future versions of the specification.

Endianness

An aspect of the system memory mapping.

See also Big-endian memory and Little-endian memory.

Hit

See Cache hit.

IMPLEMENTATION DEFINED

Means that the behavior is not defined by this specification, but must be defined and documented by individual implementations.

In a timely manner

The protocol cannot define an absolute time within which something must occur. However, in a sufficiently idle system, it will make progress and complete without requiring any explicit action.

Initiating master

A master that issues a transaction that starts a sequence of events. When describing a sequence of transactions, the term initiating master distinguishes the master that triggers the sequence of events from any snooped master that is accessed as a result of the action of the initiating master.

Initiating master is a temporal definition, meaning it applies at particular points in time, and typically is used when describing sequences of events. A master that is an initiating master for one sequence of events can be a snooped master for another sequence of events.

See also Caching master, Downstream cache, Local cache, Peer cache, Snooped master.

Interconnect component

A component with more than one AMBA interface that connects one or more master components to one or more slave components

An interconnect component can be used to group together either:

- a set of masters so that they appear as a single master interface
- a set of slaves so that they appear as a single slave interface.

See also Component, Master component, Slave component.

Line

See Cache line.

Little-endian memory

Means that the *least significant byte* (LSB) of the data is stored in the memory location with the lowest address.

See also Big-endian memory, Endianness.

Load

The action of a master component reading the value held at a particular address location. For a processor, a load occurs as the result of executing a particular instruction. Whether the load results in the master issuing a read transaction depends on whether the accessed cache line is held in the local cache.

See also Caching master, Speculative read, Store.

Local cache

A local cache is defined from the perspective of an initiating master. A local cache is one that is internal to the master. Any access to the local cache is performed within the master.

See also Initiating master, Peer cache.

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Main memory The memory that holds the data value of an address location when no cached copies of that location exist. For any location, main memory can be out of date with respect to the cached copies of the location, but main memory is updated with the most recent data value when no cached copies exist.

Main memory can be referred to as memory when the context makes the intended meaning clear.

Master component

A component that initiates transactions.

It is possible that a single component can act as both a master component and as a slave component. For example, a *Direct Memory Access* (DMA) component can be a master component when it is initiating transactions to move data, and a slave component when it is being programmed.

See also Component, Master component, Slave component.

Memory barrier *See* Barrier.

Memory Management Unit (MMU)

Provides detailed control of the part of a memory system that provides address translation. Most of the control is provided using translation tables that are held in memory, and define the attributes of different regions of the physical memory map.

See also System Memory Management Unit (SMMU).

Memory slave component

A memory slave component, or *memory slave*, is a slave component with the following properties:

- a read of a byte from a memory slave returns the last value written to that byte location
- a write to a byte location in a memory slave updates the value at that location to a new value that is obtained by subsequent reads
- reading a location multiple times has no side-effects on any other byte location
- reading or writing one byte location has no side effects on any other byte location.

See also Component, Master component, Peripheral slave component.

Miss *See* Cache miss.

MMU *See* Memory Management Unit (MMU).

Observer A processor or other master component, such as a peripheral device, that can generate reads from or writes to memory.

Peer cache A peer cache is defined from the perspective of an initiating master. A peer cache for that master is one that is accessed using the snoop channels. An initiating master cannot allocate cache lines into a peer cache.

See also Initiating master, Local cache.

Peripheral slave component

A peripheral slave component is also described as a *peripheral slave*. This specification recommends that a peripheral slave has an IMPLEMENTATION DEFINED method of access that is typically described in the data sheet for the component. Any access that is not defined as permitted might cause the peripheral slave to fail, but must complete in a protocol-correct manner to prevent system deadlock. The protocol does not require continued correct operation of the peripheral.

In the context of the descriptions in this specification, peripheral slave is synonymous with *peripheral*, *peripheral component*, *peripheral device*, and *device*.

See also Memory slave component, Slave component.

Permission to store

A master component has permission to store if it can perform a store to the associated cache line without informing any other caching master or the interconnect.

See also Caching master, Master component, Permission to update main memory, Store.

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Permission to update main memory

A master component has permission to update main memory if the master can perform a write transaction to main memory. The ACE protocol ensures that no other master performs a write transaction to the same cache location at the same time.

See also Caching master, Master component, Main memory, Permission to store, Store.

Prefetching

Prefetching refers to speculatively fetching instructions or data from the memory system. In particular, instruction prefetching is the process of fetching instructions from memory before the instructions that precede them, in simple sequential execution of the program, have finished executing. Prefetching an instruction does not mean that the instruction has to be executed.

In this manual, references to instruction or data fetching apply also to prefetching, unless the context explicitly indicates otherwise.

Slave component

A component that receives transactions and responds to them.

It is possible that a single component can act as both a slave component and as a master component. For example, a *Direct Memory Access* (DMA) component can be a slave component when it is being programmed and a master component when it is initiating transactions to move data.

See also Master component, Memory slave component, Peripheral slave component.

Snooped cache

A hardware-coherent cache on a snooped master. That is, it is a hardware-coherent cache that receives snoop transactions.

The term snooped cache is used in preference to the term snooped master when the sequence of events being described only involves the cache and does not involve any actions or events on the associated master.

See also Snooped master,

Snoop filter

A precise snoop filter that is able to track precisely the cache lines that might be allocated within a master.

Snooped master

A caching master that receives snoop transactions.

Snooped master is a temporal definition, meaning it applies at particular points in time, and typically is used when describing sequences of events. A master that is a snooped master for one sequence of events can be an initiating master for another sequence of events.

See also Caching master, Initiating master, Snooped cache.

Speculative read 预取

A transaction that a master issues when it might not need the transaction to be performed because it already has a copy of the accessed cache line in its local cache. Typically, a master issues a speculative read in parallel with a local cache lookup. This gives lower latency than looking in the local cache first, and then issuing a read transaction only if the required cache line is not found in the local cache.

See also Caching master, Load.

Store

The action of a master component changing the value held at a particular address location. For a processor, a store occurs as the result of executing a particular instruction. Whether the store results in the master issuing a read or write transaction depends on whether the accessed cache line is held in the local cache, and if it is in the local cache, the state it is in.

See also Caching master, Load, Permission to update main memory, Permission to store.

Synchronization barrier

See Barrier.

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System Memory Management Unit (SMMU)
A system-level MMU. That is, a system component that provides address translation from a one address space to another. An SMMU provides one or more of:

- *virtual address* (VA) to *physical address* (PA) translation
- VA to *intermediate physical address* (IPA) translation
- IPA to PA translation.

TLB *See* Translation Lookaside Buffer (TLB).

Transaction An AXI master initiates an AXI transaction to communicate with an AXI slave. Typically, the transaction requires information to be exchanged between the master and slave on multiple channels. The complete set of required information exchanges form the AXI transaction.

See also Beat, Burst.

Translation Lookaside Buffer (TLB)
A memory structure containing the results of translation table walks. TLBs help to reduce the average cost of a memory access.

See also System Memory Management Unit (SMMU), Translation table, Translation table walk.

Translation table
A table held in memory that defines the properties of memory areas of various sizes from 1KB.

See also Translation Lookaside Buffer (TLB), Translation table walk.

Translation table walk
The process of doing a full translation table lookup.

See also Translation Lookaside Buffer (TLB), Translation table.

Unaligned An unaligned access is an access where the address of the access is not aligned to the size of an element of the access.

Unaligned memory accesses
Are memory accesses that are not, or might not be, appropriately halfword-aligned, word-aligned, or doubleword-aligned.

See also *Aligned* on page Glossary-301

UNPREDICTABLE
In the AMBA AXI and ACE Architecture means that the behavior cannot be relied upon.

UNPREDICTABLE behavior must not be documented or promoted as having a defined effect.

Upstream An AXI transaction operates between a master component and one or more slave components, and can pass through one or more intermediate components. At any intermediate component, for a given transaction, *upstream* means between that component and the originating master component, and includes the originating master component.

Downstream and upstream are defined relative to the transaction as a whole, not relative to individual data flows within the transaction.

See also Downstream, Master component, Slave component.

Write-Back cache
A cache in which when a cache hit occurs on a store access, the data is only written to the cache. Data in the cache can therefore be more up-to-date than data in main memory. Any such data is written back to main memory when the cache line is cleaned or re-allocated. Another common term for a Write-Back cache is a *copy-back cache*.

Write-Through cache
A cache in which when a cache hit occurs on a store access, the data is written both to the cache and to main memory. This is normally done via a write buffer, to avoid slowing down the processor.

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